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RESEARCH INTERESTS

- **Identifying rare signal in complex data** — identifying rare astrophysical signals in high-dimensional, background-dominated datasets collected by different observatories, directly analogous to anomaly detection and rare event detection in industry settings.
- **ML-driven signal extraction and classification** — developing and evaluating ML models to improve event classification accuracy, energy reconstruction, and angular resolution in gamma-ray data; applicable to any high-stakes classification problem with imbalanced, noisy labels.
- **Multi-Instrument data Integration and joint inference** — building statistical frameworks that combine heterogeneous datasets from multiple instruments into a unified likelihood model (i.e. 3ML); directly transferable to multi-modal data integration.
- **End-to-end scientific data pipelines** — designing automated, scalable workflows for ingestion, calibration, and analysis of terabyte-scale telescope data on distributed HPC systems; applicable to production data engineering and MLOps.
- **Retrieval-Augmented Generation (RAG) for scientific literature** — designing and building an end-to-end RAG pipeline over 44,000+ astrophysics paper abstracts from arXiv, encompassing dense embedding generation, semantic vector indexing, query-driven retrieval, and context-augmented response generation using instruction-tuned LLMs; directly transferable to enterprise knowledge retrieval, document Q&A systems, and LLM-powered search applications.

EDUCATION

- **Ph.D.**, "*Study of the very high energy gamma-ray emission from supernova remnants with VERITAS. Two test cases: Cassiopeia A and IC 443*", Advisor: **Jamie Holder**, The University of Delaware, USA — May 2018
- **M.Tech** (Instrumentation), Panjab University, Chandigarh, India — June 2010
- **M.Sc** (Physics, Honours), Panjab University, Chandigarh, India — June 2008

APPOINTMENTS

- Postdoctoral Associate, Department of Physics and Astronomy, **University of Maryland**, College Park, USA 2022-2025
- Postdoctoral Researcher, Department of Physics and Astronomy, **University of Delaware**, Newark, USA 2021-2022
- Postdoctoral Researcher, Department of Physics, **McGill University**, Canada 2018-2021

FELLOWSHIPS AND SCHOLARSHIPS

- Dissertation fellowship, University of Delaware, USA 2016
- iProgress scholarship, Helmholtz Alliance for Astroparticle Physics, Germany 2015
- Graduate Aptitude Test in Engineering scholarship to pursue M.Tech, India 2008-2010
- Junior research fellowship in sciences, India 2008

GRANT AWARDED

As a Principal Investigator

- Fermi-Guest Investigator award ($70000), "Systematic study of the radio-dim, gamma-ray bright SNR population to test evolutionary model" 2024

TEACHING EXPERIENCE

- Lab Teaching Assistant, Fundamentals of Physics II, UDEL Spring 2012
- Lab Teaching Assistant, Electronics for Scientists, UDEL Fall 2011
- Lab Teaching Assistant, Fundamentals of Physics I, UDEL Spring 2011
- Examination, quiz and Lab grader, UDEL 2010-2012

ADVISING/MENTORING EXPERIENCE

- Co-adviser, Graduate student, UMD 2023-2025
- Mentor, Fermi-LAT mentoring program, UMD 2024-2025
- Mentor, Doctoral student/Postdoc peer mentoring program, UMD 2024
- Adviser, Summer Graduate Student, UMD 2022

OUTREACH AND SERVICE WORK

- Member, Diversity and Inclusion advisory council at College of Computer, Mathematical and Natural sciences, UMD 2024-2025
- Panelist, Fermi Gamma-Ray Space Telescope Guest Investigator Program April 2024
- Volunteer as Mentor, Data Science Info Challenge, UMD February 2024
- Referee, Nuclear Instruments and Methods in Physics Research February 2023
- Referee, The Astrophysical Journal January 2022

LEADERSHIP EXPERIENCE

- Co-Chair of Galactic Science Working Group in VERITAS collaboration 2020-2024
- Led VERITAS observing shifts and mentored junior graduate and undergraduate students in telescope operations 2014-2025

PRESS COVERAGE

- Astronomy and Space news, Phys.org, "Mysterious ultra-high energy source investigated by astronomers" September 2023

WORKSHOPS ATTENDED

- Faculty Teaching Institute (FTI), Focused on effective and inclusive teaching practices in physics and astronomy November 2023
- Summer School in Statistics for Astronomers XV, Penn State University June 2019

COLLABORATIONS

- Member of gamma-ray observatories including VERITAS, HAWC and Fermi-LAT
  – Covers an extensive energy range from MeV to PeV
  – Crucial for multi-messenger and time-domain astronomy

SHORT-TERM HARDWARE PROJECTS

- **Detector development:** Assisted in Electron-beam calibration of aerogel tiles for the HELIX RICH detector, Vickers electron linac at the NRC in Ottawa 2020
- **Detector safety:** Developed a front-end panel for observers in control room by utilizing pressure sensor data from chillers to safeguard the VERITAS telescopes data acquisition system 2014
- **Detector development:** Assembled and tested the photomultiplier tubes for the camera upgrade of VERITAS telescope 2011-2012

OPEN-SOURCE SOFTWARE DEVELOPMENT

- 3ML: The Multi-Mission Maximum Likelihood Framework 2022

TALKS/POSTERS

Invited Talks

- Highlights of VERITAS Galactic Science Results
  Mid-Atlantic APS meeting, USA November 2024

Contributed Talks

- Gamma-Ray Study of Supernova Remnants and Pulsar Wind Nebulae with VERITAS
  Bartol Centennial Meeting, USA September 2024
- Investigating the origin of gamma-ray emission from the unidentified PeVatron LHAASO J2108+5157 using data from VERITAS, HAWC, Fermi-LAT and XMM-Newton
  11th Fermi Symposium, USA September 2024
- Exploring the origin of galactic cosmic rays using the VERITAS gamma-ray observatory
  COSPAR, South Korea July 2024
- VERITAS and HAWC observations of unidentified source LHAASO J2108+5157
  38th ICRC, JAPAN August 2023
- Searching for TeV emission from Galactic PeVatrons with VERITAS
  CRSIG forum at NASA, USA January 2023
- Constraining the evaporation rate of primordial black holes using VERITAS archival data
  AAS 241, USA January 2023
- Supernova remnants at TeV energies with VERITAS
  TeVPA, Canada August 2022
- Understanding the origin of Galactic Cosmic rays with VERITAS
  COSPAR, Australia January 2021
- Deep VERITAS observations of Supernova remnants Cas A and IC 443
  Mid-Atlantic APS meeting, USA October 2016
- Detailed study of two well-established supernova remnants using VERITAS
  DESY, Germany November 2015
- A detailed stduy of gamma-ray emission from Cassiopeia A using VERITAS
  34th ICRC, The Netherlands, August 2015

Contributed Posters

- Application of Decision tree algorithms to classify signal and background events in Imaging Atmospheric Cherenkov telescopes
  Data Science Symposium, University of Delaware November 2021
- Constraining the evaporation rate of primordial black holes using VERITAS archival data
  36th ICRC, USA July 2019
- The TeV morphology of the interacting supernova remnant IC 443
  6th Fermi Symposium November 2015

TECHNICAL SKILLS

- **Programming Languages:** Python, Bash, SQL, R, C++ (ROOT/CERN)
- **Python Stack:** NumPy, SciPy, pandas, scikit-learn, PyTorch, Matplotlib, Seaborn, Plotly, Jupyter
- **Machine Learning:** Classification (Random Forest, Gradient Boosting, BDT), Clustering (K-means, DBSCAN, HDBSCAN), Convolutional Neural Networks, Transformer-based LLMs, Embedding models and fine-tuning, RAG pipeline design and deployment
- **Data Engineering:** Large-scale pipeline design, workflow automation, distributed HPC computing (Slurm), Amazon S3, AWS Lambda, SageMaker
- **MLOps & Infrastructure:** Docker, Apptainer, GitHub, Huggingface
- **Databases:** MySQL, vector databases (FAISS)
- **Scientific Domain:** Gamma-ray data analysis (Fermi-LAT, VERITAS, HAWC), multi-wavelength spectral modeling (keV – TeV), statistical inference and likelihood analysis

Publications

Peer Reviewed

Google scholar profile

[1] C. B. Adams et al. "HAWC, VERITAS, Fermi-LAT, and XMM-Newton Follow-up Observations of the Unidentified Ultra-high-energy Gamma-Ray Source LHAASO J2108+5157". In: *The Astrophysical Journal* 991.2, 192 (Oct. 2025), p. 192. DOI: 10.3847/1538-4357/adf439. arXiv: 2508.01934 [astro-ph.HE].

[2] A. Acharyya et al. "A Multiwavelength Study to Decipher the 2017 Flare of the Blazar OJ 287". In: *The Astrophysical Journal* 973.2, 134 (Oct. 2024), p. 134. DOI: 10.3847/1538-4357/ad64d0. arXiv: 2407.11848 [astro-ph.HE].

[3] A. Acharyya et al. "Indirect search for dark matter with a combined analysis of dwarf spheroidal galaxies from VERITAS". In: *Physical Review D* 110.6, 063034 (Sept. 2024), p. 063034. DOI: 10.1103/PhysRevD.110.063034. arXiv: 2407.16518 [astro-ph.HE].

[4] A. Acharyya et al. “Multiwavelength Investigation of γ-Ray Source MGRO J1908+06 Emission Using Fermi-LAT, VERITAS and HAWC”. In: *The Astrophysical Journal* 974.1, 61 (Oct. 2024), p. 61. DOI: 10.3847/1538-4357/ad698d. arXiv: 2408.01625 [astro-ph.HE].

[5] I. Pope et al. “A Multiwavelength Investigation of PSR J2229+6114 and its Pulsar Wind Nebula in the Radio, X-Ray, and Gamma-Ray Bands”. In: *The Astrophysical Journal* 960.1, 75 (Jan. 2024), p. 75. DOI: 10.3847/1538-4357/ad0120. arXiv: 2310.04512 [astro-ph.HE].

[6] A. Acharyya et al. “A VERITAS/Breakthrough Listen Search for Optical Technosignatures”. In: *The Astronomical Journal* 166.3, 84 (Sept. 2023), p. 84. DOI: 10.3847/1538-3881/ace347. arXiv: 2306.17680 [astro-ph.IM].

[7] A. Acharyya et al. “Multiwavelength Observations of the Blazar PKS 0735+178 in Spatial and Temporal Coincidence with an Astrophysical Neutrino Candidate IceCube-211208A”. In: *The Astrophysical Journal* 954.1, 70 (Sept. 2023), p. 70. DOI: 10.3847/1538-4357/ace327. arXiv: 2306.17819 [astro-ph.HE].

[8] A. Acharyya et al. “VERITAS and Fermi-LAT Constraints on the Gamma-Ray Emission from Superluminous Supernovae SN2015bn and SN2017egm”. In: *The Astrophysical Journal* 945.1, 30 (Mar. 2023), p. 30. DOI: 10.3847/1538-4357/acb7e6. arXiv: 2302.06686 [astro-ph.HE].

[9] Kaya Mori et al. “The high energy X-ray probe (HEX-P): Galactic PeVatrons, star clusters, superbubbles, microquasar jets, and gamma-ray binaries”. In: *Frontiers in Astronomy and Space Sciences* 10, 1303197 (Dec. 2023), p. 1303197. DOI: 10.3389/fspas.2023.1303197. arXiv: 2311.04851 [astro-ph.HE].

[10] H. Abe et al. “Gamma-ray observations of MAXI J1820+070 during the 2018 outburst”. In: *Monthly Notices of the Royal Astronomical Society* 517.4 (Dec. 2022), pp. 4736–4751. DOI: 10.1093/mnras/stac2686. arXiv: 2209.09785 [astro-ph.HE].

[11] C. B. Adams et al. “Variability and Spectral Characteristics of Three Flaring Gamma-Ray Quasars Observed by VERITAS and Fermi-LAT”. In: *The Astrophysical Journal* 924.2, 95 (Jan. 2022), p. 95. DOI: 10.3847/1538-4357/ac32bd. arXiv: 2110.13181 [astro-ph.HE].

[12] C. B. Adams et al. “An Archival Search for Neutron-star Mergers in Gravitational Waves and Very-high-energy Gamma Rays”. In: *The Astrophysical Journal* 918.2, 66 (Sept. 2021), p. 66. DOI: 10.3847/1538-4357/ac0623. arXiv: 2106.01386 [astro-ph.HE].

[13] C. B. Adams et al. “Observation of the gamma-ray binary HESS J0632+057 with the H.E.S.S., MAGIC, and VERITAS telescopes”. In: *arXiv e-prints*, arXiv:2109.11894 (Sept. 2021), arXiv:2109.11894. arXiv: 2109.11894 [astro-ph.HE].

[14] C. B. Adams et al. “VERITAS Observations of the Galactic Center Region at Multi-TeV Gamma-Ray Energies”. In: *The Astrophysical Journal* 913.2, 115 (June 2021), p. 115. DOI: 10.3847/1538-4357/abf926. arXiv: 2104.12735 [astro-ph.HE].

[15] W. Benbow et al. “A Search for TeV Gamma-Ray Emission from Pulsar Tails by VERITAS”. In: *The Astrophysical Journal* 916.2, 117 (Aug. 2021), p. 117. DOI: 10.3847/1538-4357/ac05b9. arXiv: 2105.13911 [astro-ph.HE].

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[23] A. U. Abeysekara et al. “Periastron Observations of TeV Gamma-Ray Emission from a Binary System with a 50-year Period”. In: *The Astrophysical Journal, Letters* 867.1, L19 (Nov. 2018), p. L19. DOI: 10.3847/2041-8213/aae70e. arXiv: 1810.05271 [astro-ph.HE].

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[27] A. Archer et al. “HESS J1943+213: An Extreme Blazar Shining through the Galactic Plane”. In: *The Astrophysical Journal* 862.1, 41 (July 2018), p. 41. DOI: 10.3847/1538-4357/aacbd0. arXiv: 1806.04144 [astro-ph.HE].

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[49] E. Aliu et al. "A Search for Enhanced Very High Energy Gamma-Ray Emission from the 2013 March Crab Nebula Flare". In: *The Astrophysical Journal, Letters* 781.1, L11 (Jan. 2014), p. L11. DOI: 10.1088/2041-8205/781/1/L11. arXiv: 1309.5949 [astro-ph.HE].

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Conference Proceedings

[1] S. Kumar, M. Martin, and X. Wang. "VERITAS and HAWC observations of unidentified source LHAASO J2108+5157". In: *arXiv e-prints*, arXiv:2309.00089 (Aug. 2023). DOI: 10.48550/arXiv.2309.00089. arXiv: 2309.00089.

[2] S. Kumar and VERITAS Collaboration. "Constraining the evaporation rate of Primordial black holes using archival data from VERITAS". In: *36th International Cosmic Ray Conference (ICRC2019)*. Vol. 36. International Cosmic Ray Conference. July 2019, 719, p. 719. arXiv: 1909.01171.

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In Preparation

[1] **S. Kumar**, J. Hewitt, B. Humensky *et al.*, "Deatiled spectral and morphological study of gamma-ray SNR IC 443 with VERITAS" (draft available upon request)

[2] J. Eagle, **S. Kumar**, R. Shang, T. Ramiro *et al.*, "Multiwavelength Investigation of HESS J1857+026 Emission Using the Fermi–LAT, VERITAS, and HAWC Observatories" (already submitted)

References

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